

Cross-Age Face Verification from Mined Same-Person Social-Media Pairs

Anonymous submission

Abstract

Cross-age face verification—deciding whether two photographs taken years apart depict the same person—is hard, and longitudinal training data are scarce. We ask a *data-centric* question: can supervision for age invariance be obtained without a manually labeled longitudinal dataset? We show that multi-photo “then/now” social-media posts supply weak *same-person* longitudinal pairs with no manual identity or age labels (a large language model parses captions and embedding clustering forms persons; we audit these components, including a small manual validation), and that fine-tuning a deliberately weak recognizer on them induces *transferable, targeted* large-gap robustness. On the external benchmark FG-NET, large-gap (≥ 25 -year) ROC-AUC rises **from 0.736 to 0.848** ($+0.112 \pm 0.001$ over three seeds; DeLong $p < 10^{-19}$), at the cost of -0.019 LFW accuracy and *substantial* low-FAR degradation—a targeted retrieval gain, not a general-verifier upgrade. Our central, bounded claim is that, within this weak-backbone attribution protocol, the *data source*—not the choice among the objectives we test—accounts for most of the gain: a pair-contrastive objective matches an ArcFace-margin one (0.848 vs. 0.851). The gain is driven by the same-person pairing (caption age anchors add only $+0.009$), is mechanistically tied to removing an age shortcut, beats a real re-aging model under a gap-parity control, and transfers to an independent non-VK platform.

1 Introduction

Cross-age face verification arises in archival retrieval, account recovery, and authorized search for missing persons. The challenge lies not in discriminating among arbitrary individuals but in distinguishing *the same person across age* from *different people of similar age and appearance*. Annotating longitudinal identity pairs (one individual across many years) is costly, and such data remain scarce, which constrains supervised approaches; recent work continues to report weak cross-dataset generalization and a shortage of high-quality longitudinal data.

Hypothesis. A multi-photo “then/now” post showing one person at different ages yields $\binom{n}{2}$ positive identity pairs without manual annotation; a caption stating ages adds a

This is an anonymized submission for review purposes only. Distribution, citation, or public sharing of this manuscript is strictly prohibited. Copyright and publication details will appear in the final version if accepted.

secondary age anchor. The same-person pairing is the operative signal (an ablation isolates the age anchor at a marginal $+0.009$); this is a scalable, weakly/naturally-supervised signal of cross-age identity, and fine-tuning a weak recognizer on it should transfer to external benchmarks.

Contributions (bounded). (1) A weakly/naturally-supervised (no manual id/age labels) longitudinal signal mined from social posts—same-person identity pairs, with caption age anchors only a secondary cue. (2) A rigorous *attribution protocol* (one weak trainable backbone, frozen vs. fine-tuned, external evaluation) that isolates the contribution of *data* from network capacity. (3) Mutually reinforcing controls (loss family, architecture, source community, training objective, real-vs-synthetic, shortcut diagnosis, cross-platform transfer) supporting the bounded claim that, within this protocol and across the objectives tested, the data contributes more to the large-gap gain than the choice among loss families. (4) A curation pipeline with an audit of its automatic components, and the finding that naive positive counts are inflated about $2\times$ by near-duplicate frames. The novelty is data-centric rather than architectural—a new *origin of supervision*—and should be evaluated accordingly. A supplementary technical appendix reports the controls and audits deferred for space.

2 Related Work and Positioning

Three lines of prior work are most relevant. *Discriminative age-invariant FR on labeled data*: OE-CNN (orthogonal embedding decomposition) (Wang et al. 2018), DAL (decorrelated adversarial learning) (Wang et al. 2019), and MTLFace (multi-task recognition, age synthesis and attention decomposition) (Huang, Zhang, and Shan 2021) achieve strong accuracy but are trained on datasets with *manual identity and age labels* (CACD (Chen, Chen, and Hsu 2014), MORPH (Ricanek and Tesafaye 2006), AgeDB (Moschoglou et al. 2017), FG-NET (Panis et al. 2016)). *Cross-age contrastive with synthetic samples*: CACon (Wang, Sanchez, and Li 2024) adds a synthesized cross-age sample and a triplet loss; OrdCon (Wang, Sanchez, and Li 2025) uses order-enhanced contrastive learning. *Synthetic aging for robustness*: systematic studies show synthetic aging helps only partially (Yao et al. 2024). Generic self-supervised FR (SimCLR-style augmentation (Chen et al. 2020)) lacks identity anchors across time. We instead mine the structure

“one post = one person at different ages” as a label-free positive-pair generator (a per-method positioning table is in the supplement). We report on the canonical cross-age benchmarks of these methods (AgeDB-30, CALFW (Zheng, Deng, and Hu 2017), FG-NET; CALFW is a harder, age-gapped LFW (Huang et al. 2007)). The goal is *not* to surpass the absolute state of the art (we use a weak backbone for clean attribution) but to quantify the magnitude and transferability of the gain from a naturally-supervised source; Sec. 5.1 trains the canonical SOTA objective on our data directly.

3 Data: Naturally-Anchored Longitudinal Pairs

Source. Two public “then/now” multi-photo communities on VK (one platform, Russian-language context). Faces are detected and aligned with a 5-point ArcFace template (InsightFace RetinaFace (Deng et al. 2020b), `buffalo_1`). A *usable* face passes detection-score, minimum-size, blur, and single-target checks (Sec. 4). The raw mined corpus is summarized in Table 1.

Age extraction. Captions are parsed by regular expressions (Russian patterns plus the slash format “6/18/21”) and by a large language model (LLM). Across all 33,697 multi-photo captions the LLM broadens and regularizes extraction relative to regex—it avoids reading calendar years or gaps as ages and recovers second ages—disagreeing on 27.2% of posts, overwhelmingly LLM-favoring; on a 100-caption manual gold subset the LLM matched the human reading on 89% vs. regex 71% (supplement). We do not treat LLM output as ground truth: downstream, explicit age anchors contribute only marginally (Sec. 5.1).

Persons and leakage-safe split. A naive “one post = one identity” undercounts: a person recurs across posts. We cluster post-groups into persons by embedding similarity (cosine ≥ 0.85), giving 21,848 persons from 27,487 post-groups ($\approx 20\%$ recurrence), and split *by person* (not by post). Manual inspection showed cosine does not separate look-alikes from same-person in the 0.55–0.8 band, so the merge threshold is deliberately conservative.

Auditing the supervision. The automatic components are potential hidden label noise, audited three ways: LLM-vs-regex age agreement; within-person apparent-gender consistency as an inexpensive multi-person detector; and a *small* manual validation (one annotator) that *sanity-checks* the pipeline (age, group integrity, pair and merge precision 0.93–0.96; supplement). With a single annotator and no inter-annotator κ this *supports* but does not *certify* the supervision; a large multi-annotator audit is beyond our resources, so residual noise is acknowledged as a limitation (Sec. 7).

Curation and its largest finding. We apply per-person apparent gender/age aggregation, LLM group-integrity validation (92.8% single-person), and near-duplicate dedup within each person (cosine ≥ 0.97). Because the number of positives is quadratic in faces per group, removing near-duplicates cut the positive count from 65,897 to 31,865 (–52%); **about half of the raw “positives” were duplicate frames** (a single shoot, burst, or repost) inflating internal metrics. After curation the frozen internal baseline drops

(our.25 + 0.705 $\rightarrow$ 0.640), exposing a harder, more representative benchmark; external benchmarks are unaffected. The reduction is stable across dedup thresholds 0.93–0.97 (supplement). Provenance, rights and governance fields are stored per item; a full datasheet (Geburu et al. 2021) accompanies the controlled-access release (see the Ethical Statement).

4 Method, Protocol and Statistics

Attribution protocol. Comparing an adapter on top of a *frozen* strong ArcFace against frozen ArcFace is invalid (the adapter merely deforms near-perfect embeddings). Instead we use *one weak trainable backbone* (FaceNet InceptionResnetV1 (Schroff, Kalenichenko, and Philbin 2015), CASIA-WebFace) and compare (a) frozen $\rightarrow$ benchmark metric vs. (b) soft fine-tuning of the head on our pairs $\rightarrow$ benchmark metric, with a margin contrastive loss on aligned crops. *Evaluation is on external benchmarks* (LFW, AgeDB-30, CALFW, FG-NET) for a clean attribution of transfer; the internal test (our. *) is a secondary in-domain diagnostic. Backbone strength is varied (Sec. 5.3) to map where the data helps. The design trades absolute performance for *causal attribution*.

Backbones and metrics. FaceNet (weak); ArcFace iResNet-50 (Deng et al. 2019) (weak, different architecture); AdaFace IR-50/IR-101 (Kim, Jain, and Liu 2022) and ArcFace iResNet-100 (strong; a clean depth control). We report LFW (10-fold accuracy), AgeDB-30 and CALFW (ROC-AUC and accuracy), and FG-NET (ROC-AUC overall and on the large-gap ≥ 25 -year subset), distinguishing *threshold-free* metrics (ROC-AUC) from *threshold-based* ones (accuracy, TAR@FAR).

Statistics. The *primary endpoint* is FG-NET large-gap ROC-AUC, with the null hypothesis that fine-tuning does not improve it over frozen (gain ≤ 0); AgeDB-30, CALFW and the internal cross-age test are secondary, and easy LFW accuracy is a forgetting control. The headline gain is mean $\pm$ std over three seeds. We report bootstrap CIs, EER and TAR@FAR (the internal test additionally uses an identity-level, person-resampling bootstrap), DeLong’s test for two *correlated* ROC curves (DeLong, DeLong, and Clarke-Pearson 1988), a paired permutation test, and a Holm correction across the stratified fairness tests (full operating-point tables in the supplement). All headline experiments use the curated data (21,427 groups, 40,586 faces, 31,865 positives; by-person split with balanced negatives).

5 Results

5.1 Main Effect and Supervision-Source Ablation

The weak FaceNet, softly fine-tuned on our pairs, raises FG-NET large-gap ROC-AUC **0.736 $\rightarrow$ 0.848** ($+0.112 \pm 0.001$), while easy LFW falls only -0.019 and AgeDB-30/CALFW move negligibly (Table 2). An ablation isolates the source: same-post pairs use no manual labels and deliver almost all of the gain; adding explicit age anchors contributes only **+0.009** on the 25+ bucket. The method is therefore *weakly supervised by naturally occurring same-post co-occurrence*, with age supervision marginal.

Stage	Posts	Photos	Usable faces	Persons (groups)	Positive pairs
Raw mined (2 communities)	44,609	84,147	49,606	21,848	65,897
After curation	—	—	40,586	21,427	31,865

Table 1: Dataset scale (raw mined) and after curation. Removing near-duplicate frames roughly halves the positive count.

Metric	Frozen	+pairs (mean $\pm$ std)	Gain	(Pre-clean gain)
FG-NET large-gap	0.7361	0.8484 $\pm$ 0.0007	+0.1124	+0.1215
FG-NET ROC	0.8955	0.9230 $\pm$ 0.0006	+0.0275	+0.0277
our.25+ (internal)	0.6403	0.8348 $\pm$ 0.0027	+0.1946	+0.1677
our.overall (internal)	0.8363	0.9163 $\pm$ 0.0009	+0.0801	+0.0764
LFW accuracy	0.9688	0.9503 $\pm$ 0.0031	−0.0185	−0.0283
AgeDB-30 ROC	0.9530	0.9462 $\pm$ 0.0013	−0.0068	−0.0155
CALFW ROC	0.9482	0.9489 $\pm$ 0.0014	+0.0007	−0.0030

Table 2: Main result: weak FaceNet frozen vs. +pairs, three seeds, curated data (pre-clean gain shown for reference). The gain is concentrated on the FG-NET large-gap subset.

Bootstrap confidence intervals (seed 42) confirm the primary endpoint: FG-NET large-gap rises from 0.736 [0.70, 0.78] to 0.848 [0.82, 0.88] with *non-overlapping* 95% CIs; the equal-error rate nearly halves (0.335→0.219) and TAR@FAR=1% nearly doubles (0.21→0.39). Because bootstrap-CI overlap ignores the samples the two models share, we confirm the endpoint with DeLong’s test for correlated ROC curves (DeLong, DeLong, and Clarke-Pearson 1988): on FG-NET large-gap $z = 9.3$, $p = 1.4 \times 10^{-20}$ (ΔAUC 95% CI [+0.088, +0.135]); a 10^4 -fold paired permutation test agrees ($p < 10^{-4}$). The same test shows the gain is *targeted*: AgeDB-30 decreases slightly (−0.008, $p = 4 \times 10^{-6}$) and CALFW is unchanged ($p = 0.95$). Operating points (Table 3) make the trade-off concrete: large-gap AUC and TAR@1% rise sharply, while easy-benchmark low-FAR points (LFW TAR@0.1%) degrade.

Objective invariance. Training the canonical modern-FR margin objectives—ArcFace, CosFace and SphereFace identity classification (Deng et al. 2019) (the core of OE-CNN/MTLFace)—on our naturally-supervised identities, same backbone and protocol, the contrastive, ArcFace and CosFace objectives **coincide** on the primary endpoint (0.848 / 0.851 / 0.843, within ± 0.006); only SphereFace, hardest to optimize on 2–3-shot identities without λ -annealing, trails (0.801). A noise-robust sub-center ArcFace (Deng et al. 2020a) also lands at 0.851—explicit label-noise machinery adds nothing on the *curated* pairs. We draw the bounded conclusion that, within this protocol and among the objectives tested, the *data source* accounts for more of the gain than the objective choice (Table 4).

5.2 Real Pairs Outperform Synthetic Aging

Replacing real positives with synthetically aged versions—a domain proxy and a real re-aging network (FRAN (Zoss et al. 2022))—underperforms real pairs, and at FRAN’s default *wide-gap* setting even *degrades* below frozen (FG-NET large-gap 0.727; Table 5). Identity-preserving aging reproduces texture artifacts rather than genuine longitudinal variation (pose, camera, era). A native-resolution control rules out

a crop artifact: re-cropping a train subset at 512 px directly from the raw images and re-aging leaves the result unchanged (0.771 vs. 0.778). A gap-parity control sharpens this: FRAN ages to a wide 33–70-year gap whereas real pairs have a median gap of 9 years; matching the synthetic gap to the real distribution lifts FRAN to **0.759**—a modest positive in line with reported synthetic-aging gains (Yao et al. 2024)—yet real pairs (0.848) still lead by +0.089. Real mined pairs beat even a *gap-matched* re-aging model.

5.3 Headroom and Data Scaling

The gain reproduces on FaceNet, ArcFace iResNet, and AdaFace IR-50/IR-101 (three loss families, two architectures) and *monotonically decreases with frozen strength* (Fig. 1); a clean depth control (AdaFace IR-50 → IR-101) confirms it. Strong saturated backbones do not improve and forget slightly under fine-tuning; a gentler learning rate halves the forgetting but does not convert it into a gain. The contribution is therefore specific to weak and medium backbones with cross-age headroom—a limitation we state explicitly (Sec. 7). Varying the fraction of training identities (val/test fixed), the gain saturates early: **≈96% of the full FG-NET large-gap gain is reached with only 10% of identities** ($\approx 1,500$), plateauing from 25% (Fig. 2; each point mean $\pm$ std over three seeds). The signal is data-efficient; further gains require *harder* data (larger gaps, hard positives) rather than merely more volume.

5.4 Mechanism: the Age Shortcut

Under age-matched negatives (same age bucket, different people), frozen accuracy on the internal 25+ test falls from 0.640 to 0.517 (**near the chance level of 0.5**), and real pairs restore it to **0.838** (Fig. 3)—direct evidence that frozen models discriminate cross-age pairs largely by *age*, whereas our pairs induce discrimination by *identity*. A linear probe refines this representationally: apparent age stays well decodable from the identity embedding for *all* models (≈ 0.63 –0.65 balanced accuracy $\gg 0.25$ chance) and changes only

Benchmark	Frozen AUC [95% CI]	+pairs AUC [95% CI]	EER (fr.→+p.)	TAR@1%	TAR@0.1%
FG-NET large-gap	0.736 [0.70, 0.78]	0.848 [0.82, 0.88]	0.335→0.219	0.209→0.386	0.074→0.154
FG-NET overall	0.896 [0.89, 0.90]	0.922 [0.92, 0.93]	0.186→0.151	0.502→0.562	0.315→0.308
AgeDB-30	0.953 [0.95, 0.96]	0.945 [0.94, 0.95]	0.115→0.126	0.525→0.510	0.285→0.309
CALFW	0.948 [0.94, 0.95]	0.948 [0.94, 0.95]	0.116→0.117	0.580→0.624	0.213→0.320
LFW (AUC)	0.994 [0.99, 1.00]	0.987 [0.98, 0.99]	0.032→0.052	0.940→0.869	0.858→0.575

Table 3: Operating points and 95% bootstrap CIs (FaceNet frozen vs. +pairs, seed 42); the internal test additionally uses an identity-level bootstrap (text). LFW is shown as ROC-AUC for consistency.

Objective	FG-NET l.g.	our.25+	LFW	AgeDB-30
Frozen	0.736	0.640	0.969	0.953
+pairs (contrastive)	0.848	0.838	0.946	0.945
+ArcFace	0.851	0.868	0.952	0.947
+ArcFace, sub-center	0.851	0.869	0.951	0.947
+CosFace	0.843	0.860	0.952	0.945
+SphereFace	0.801	0.699	0.950	0.943

Table 4: Objective comparison (FaceNet, curated): contrastive, ArcFace and CosFace coincide on the FG-NET large-gap primary endpoint; a noise-robust sub-center ArcFace adds nothing; SphereFace (no λ -annealing) trails on sparse few-shot identities.

Metric	Frozen	+real	syn-FRAN	FRAN-gm
FG-NET large-gap	0.736	0.848	0.727	0.759
our.25+	0.640	0.838	0.549	0.622

Table 5: Real vs. synthetic positives (FaceNet, curated). FRAN-gm matches the synthetic age-gap to the real-pair distribution (median 9 y).

slightly, while identity AUC rises sharply ($0.836 \rightarrow 0.916$)—the method learns an invariant *metric* (the cosine stops relying on the age axis), not an age-free representation; explicit adversarial disentanglement reduces leakage no further (supplement).

5.5 Cross-Platform and Cross-Source Transfer

Training on one VK community and testing on the held-out other transfers the gain (external FG-NET large-gap +0.113, essentially identical to the within-source +0.112). A stronger external check is whether the *source* generalizes beyond its platform. Applying the *identical* mining protocol to an independent, non-VK community—the Reddit board r/PastAndPresentPics (same “then/now” format, same LLM parser)—and evaluating the VK-trained +pairs model with *no* Reddit data in training on 1,575 Reddit pairs raises overall ROC-AUC **0.718** $\rightarrow$ **0.749** (+0.031, near-disjoint 95% CIs). The transfer is *bidirectional*: a backbone trained on *only* a small 220-person Reddit source (a pilot, not a headline) also helps on VK (FG-NET large-gap +0.047). The gain is smaller than within-VK (a noisier, partly collage-derived set), so we read it as confirming the source generalizes beyond its platform, not as a SOTA claim. Figure 4 shows the +pairs gain stays *positive across all three* external evalua-

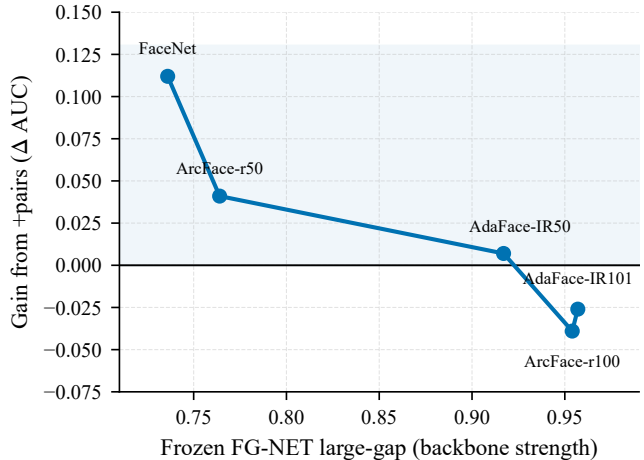


Figure 1: Headroom curve: the +pairs gain on FG-NET large-gap shrinks monotonically as the frozen backbone strengthens, crossing zero for strong saturated models.

tions, shrinking on harder/noisier sets.

Apparent-demographic audit. Stratifying by *apparent* gender/age of the pair anchor (estimated, not self-identified), no group degrades and every gain is significant (95% paired-bootstrap CI > 0; supplement). The largest gain is for the youngest group 0–17 ($0.750 \rightarrow 0.880$, +0.130)—the method strengthens where the base model is weakest—corroborated on an *independent* child↔adult FG-NET subset (+0.129, non-overlapping CIs). At a matched $\approx 1\%$ FMR, +pairs lowers the false-non-match rate in every apparent stratum. We avoid the word “fair” as a solved property: the source is apparent-female-skewed (76.7%) and attributes are apparent, not legal categories.

6 Discussion

The contribution is a scalable, weakly/naturally-supervised source of longitudinal supervision that teaches transferable cross-age robustness which synthetic aging cannot replace and explicit age supervision does not explain. Reinforcing controls—loss family, architecture, source community, objective, and a noise-robust margin—converge on a bounded conclusion: within this protocol, the *data source* accounts for more of the large-gap gain than the choice of objective. Crucially, the improvement is a *targeted* large-gap gain, not a drop-in upgrade to a general verifier: low-FAR operating points on easy benchmarks degrade (LFW

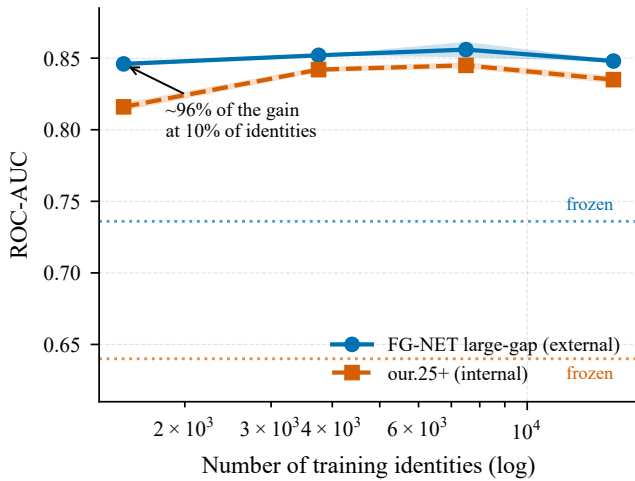


Figure 2: Data-scaling law: external (FG-NET large-gap) and internal (our.25+) ROC-AUC vs. training identities (log scale). About 96% of the gain is reached at 10% of identities; shaded bands are ± 1 std over three seeds, dashed lines the frozen baselines.

TAR@FAR=0.1% falls $0.858 \rightarrow 0.575$), so deployment should be scoped to the large-gap cross-age regime.

Scope of the claims. *Established (within this attribution protocol):* (i) the mined pairs improve large-gap cross-age verification on FG-NET (primary endpoint, non-overlapping 95% CIs) and the internal 25+ test; (ii) the gain transfers across the two source communities, to an independent child $\leftrightarrow$ adult FG-NET subset, and to a non-VK platform; (iii) real pairs **outperform synthetic aging**, including under native-resolution and gap-parity controls; (iv) the gain is invariant across the contrastive, ArcFace and CosFace objectives; (v) it is mechanistically tied to **removing an age shortcut**; (vi) roughly half of the raw positive pairs were near-duplicate frames. *Not established / out of scope:* the effect is not a uniform verification gain (low-FAR operating points degrade); strong saturated backbones do not improve; a larger non-VK *training* source with multi-seed CIs remains future work; we do not reproduce a full SOTA pipeline, only its shared margin objective; the fairness audit uses apparent attributes only and lacks a multi-annotator κ .

7 Limitations

- **Backbone dependence:** the gain concentrates in weak/medium backbones; strong saturated models do not improve and slightly forget.
- **External validity:** two VK communities, one platform, one language/cultural context; apparent-female-skewed (76.7%); sparse at apparent age 45+. Transfer is shown between two VK sources, to standard external benchmarks, and to a small independent Reddit then/now test; this does not establish platform-agnostic generalization.
- **Apparent attributes only** in the fairness audit (estimated, not self-identified gender/ethnicity), and no human-audited inter-annotator κ (a large manual audit

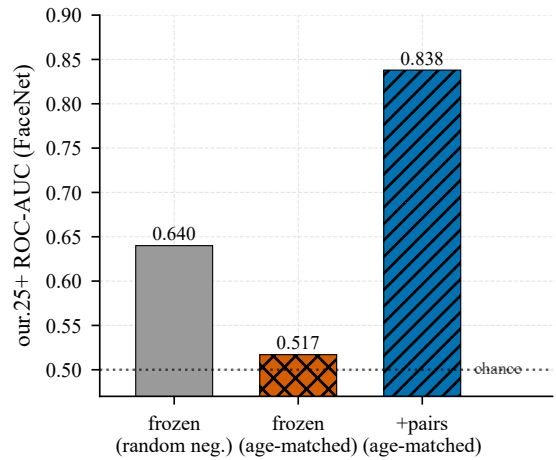


Figure 3: The age shortcut: on an identity-only test (age-matched negatives) the frozen model falls to near-chance; fine-tuning on our pairs restores discrimination by identity.

is labor-intensive and beyond our resources).

- **Weakly supervised, not label-free:** an LLM parses ages and validates group integrity and clustering forms persons—audited but a residual hidden-noise source.
- **Benchmark scale and age:** the field-standard cross-age suite (AgeDB-30, CALFW, FG-NET) fixes comparability but inherits limited scale/age; a larger non-VK longitudinal set remains useful future work.

Ethical Statement

This is a sensitive biometric study; we treat governance as a first-class component. Permitted use is restricted to controlled / consent-based / human-reviewed applications (rights-cleared archives and family photos, authorized humanitarian search, account recovery); the system outputs candidates for human review, never an automatic decision. This is a *research-only* study—nothing is deployed and *no public biometric database is released*. Forbidden: mass identification, de-anonymization, surveillance, and any processing of minors’ data without a legal basis.

Ethical approval. This study was reviewed and approved by the authors’ institutional research ethics committee; the committee name, protocol number and approval date are withheld here to preserve anonymity and will be provided in the camera-ready version. **Consent.** Because the data are drawn from public posts with no feasible channel to contact subjects, individual informed consent for participation and publication was not obtained; we rely on a scientific-research legal basis with the de-identification safeguards below, and note that official API access and voluntary public posting do not by themselves authorize biometric processing (cf. GDPR Art. 9 special-category data (European Parliament and Council of the European Union 2016) and national biometric-consent law). We report this gap transparently rather than claiming full compliance.

De-identification and release. We do not release raw face images or raw posts. The public release is restricted to

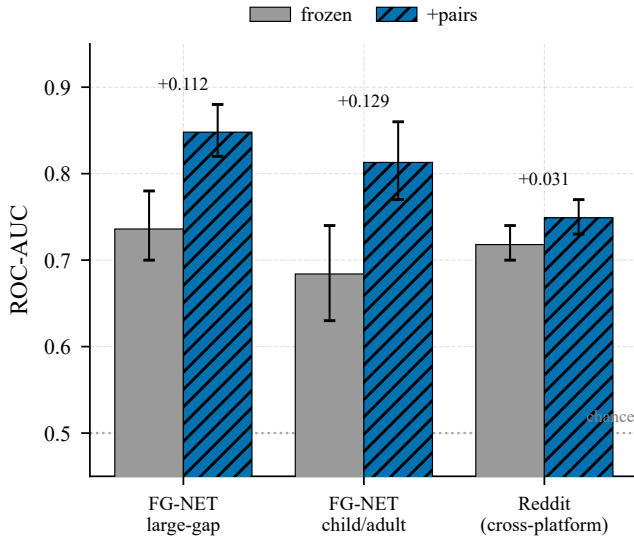


Figure 4: External generalization. The +pairs gain over frozen (95% CIs) holds across three independent evaluations—FG-NET large-gap, a child↔adult FG-NET subset, and the cross-platform Reddit set—each well above chance, largest on the in-protocol FG-NET sets and smaller (but positive) on the noisier Reddit set.

code, configurations, the pseudonymized benchmark protocol and aggregate statistics. Because a face embedding is itself a comparable *biometric template* and salted identifier hashes do not render it anonymous, trained weights and derived embeddings are made available *only* under controlled access (Data Use Agreement; research-only; no re-identification). Platform identifiers are replaced by salted hashes; captions/comments are excluded; only an age bucket and a pseudonymous person ID are kept. Minor-age items are withheld from any release absent an explicit legal framework and verifiable guardian consent. A Data Protection Impact Assessment, a full datasheet (Gebru et al. 2021) and a take-down / deletion-on-request procedure accompany the release.

Reproducibility. We release all scripts, configurations and versioned LLM prompts; raw face images and posts are withheld, so data-mining replication is partial (we release cached LLM outputs for deterministic replay). The entire study runs on a single laptop (one 6 GB GPU). A reproducibility checklist accompanies the submission.

References

Chen, B.-C.; Chen, C.-S.; and Hsu, W. H. 2014. Cross-Age Reference Coding for Age-Invariant Face Recognition and Retrieval. In *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 768–783.

Chen, T.; Kornblith, S.; Norouzi, M.; and Hinton, G. 2020. A Simple Framework for Contrastive Learning of Visual Representations. In *Proc. Int. Conf. Mach. Learn. (ICML)*, 1597–1607.

DeLong, E. R.; DeLong, D. M.; and Clarke-Pearson, D. L. 1988. Comparing the Areas under Two or More Correlated

Receiver Operating Characteristic Curves: A Nonparametric Approach. *Biometrics*, 44(3): 837–845.

Deng, J.; Guo, J.; Liu, T.; Gong, M.; and Zafeiriou, S. 2020a. Sub-center ArcFace: Boosting Face Recognition by Large-Scale Noisy Web Faces. In *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 741–757.

Deng, J.; Guo, J.; Ververas, E.; Kotsia, I.; and Zafeiriou, S. 2020b. RetinaFace: Single-Shot Multi-Level Face Localisation in the Wild. In *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 5203–5212.

Deng, J.; Guo, J.; Xue, N.; and Zafeiriou, S. 2019. ArcFace: Additive Angular Margin Loss for Deep Face Recognition. In *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 4690–4699.

European Parliament and Council of the European Union. 2016. Regulation (EU) 2016/679 (General Data Protection Regulation). Official Journal of the European Union, L119.

Gebru, T.; Morgenstern, J.; Vecchione, B.; Vaughan, J. W.; Wallach, H.; Daumé III, H.; and Crawford, K. 2021. Datasheets for Datasets. *Commun. ACM*, 64(12): 86–92.

Huang, G. B.; Ramesh, M.; Berg, T.; and Learned-Miller, E. 2007. Labeled Faces in the Wild: A Database for Studying Face Recognition in Unconstrained Environments. Technical Report 07-49, Univ. of Massachusetts, Amherst.

Huang, Z.; Zhang, J.; and Shan, H. 2021. When Age-Invariant Face Recognition Meets Face Age Synthesis: A Multi-Task Learning Framework. In *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 7282–7291.

Kim, M.; Jain, A. K.; and Liu, X. 2022. AdaFace: Quality Adaptive Margin for Face Recognition. In *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 18750–18759.

Moschoglou, S.; Papaioannou, A.; Sagonas, C.; Deng, J.; Kotsia, I.; and Zafeiriou, S. 2017. AgeDB: The First Manually Collected, In-the-Wild Age Database. In *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, 51–59.

Panis, G.; Lanitis, A.; Tsapatsoulis, N.; and Cootes, T. F. 2016. Overview of Research on Facial Ageing Using the FG-NET Ageing Database. *IET Biometrics*, 5(2): 37–46.

Ricanek, K.; and Tesafaye, T. 2006. MORPH: A Longitudinal Image Database of Normal Adult Age-Progression. In *Proc. IEEE Int. Conf. Autom. Face Gesture Recognit. (FG)*, 341–345.

Schroff, F.; Kalenichenko, D.; and Philbin, J. 2015. FaceNet: A Unified Embedding for Face Recognition and Clustering. In *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 815–823.

Wang, H.; Gong, D.; Li, Z.; and Liu, W. 2019. Decorrelated Adversarial Learning for Age-Invariant Face Recognition. In *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 3527–3536.

Wang, H.; Sanchez, V.; and Li, C.-T. 2024. Cross-Age Contrastive Learning for Age-Invariant Face Recognition. In *Proc. IEEE Int. Conf. Acoust. Speech Signal Process. (ICASSP)*. ArXiv:2312.11195.

Wang, H.; Sanchez, V.; and Li, C.-T. 2025. From Age Estimation to Age-Invariant Face Recognition: Generalized Age Feature Extraction Using Order-Enhanced Contrastive Learning. *arXiv preprint arXiv:2501.01760*.

Wang, Y.; Gong, D.; Zhou, Z.; Ji, X.; Wang, H.; Li, Z.; Liu, W.; and Mei, T. 2018. Orthogonal Deep Features Decomposition for Age-Invariant Face Recognition. In *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 738–753.

Yao, W.; Farooq, M. A.; Lemley, J.; and Corcoran, P. 2024. Synthetic Face Ageing: Evaluation, Analysis and Facilitation of Age-Robust Facial Recognition Algorithms. *arXiv preprint arXiv:2406.06932*.

Zheng, T.; Deng, W.; and Hu, J. 2017. Cross-Age LFW: A Database for Studying Cross-Age Face Recognition in Unconstrained Environments. *arXiv:1708.08197*.

Zoss, G.; Chandran, P.; Sifakis, E.; Gross, M.; Gotardo, P.; and Bradley, D. 2022. Production-Ready Face Re-Aging for Visual Effects. *ACM Trans. Graph. (SIGGRAPH Asia)*, 41(6): 1–12.